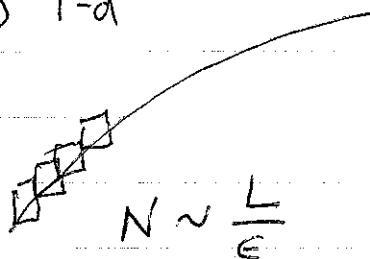


Dimension of Lorenz attractor

Box dimension: (box size ϵ)

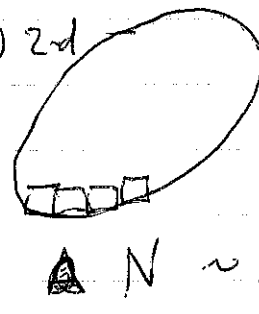
(a) 1-d



$$\ln(N) \approx \ln(L) + \ln\left(\frac{1}{\epsilon}\right)$$

$$d = \lim_{\epsilon \rightarrow 0} \frac{\ln(N(\epsilon))}{\ln\left(\frac{1}{\epsilon}\right)}$$

(b) 2-d

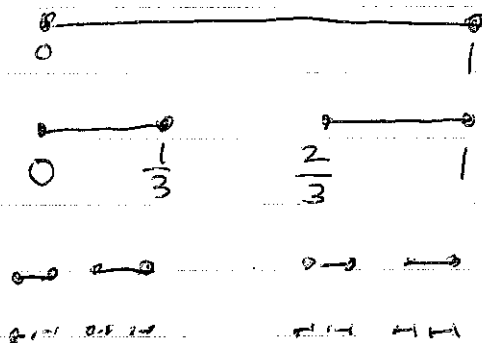


$$\ln(N) \approx \ln(A)$$

$$+ \ln\left(\frac{1}{\epsilon^2}\right)$$

$$= \ln(A) + 2 \ln\left(\frac{1}{\epsilon}\right)$$

(c) Cantor set



$$\text{let } \epsilon = \left(\frac{1}{3}\right)^n$$

$$n=0 \Rightarrow N=1$$

$$n=1 \Rightarrow N=2$$

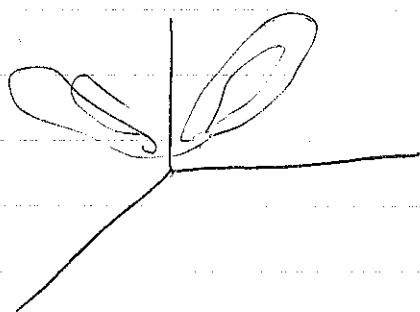
$$n=2 \Rightarrow N=4$$

$$N = 2^n$$

$$\lim_{\epsilon \rightarrow 0} = \lim_{n \rightarrow \infty}$$

$$d = \lim_{n \rightarrow \infty} \frac{\ln(2^n)}{\ln(3^n)} = \frac{\ln(2)}{\ln(3)} \approx 0.63$$

Chaotic Attractor — a measure of dimension
"correlation dimension"

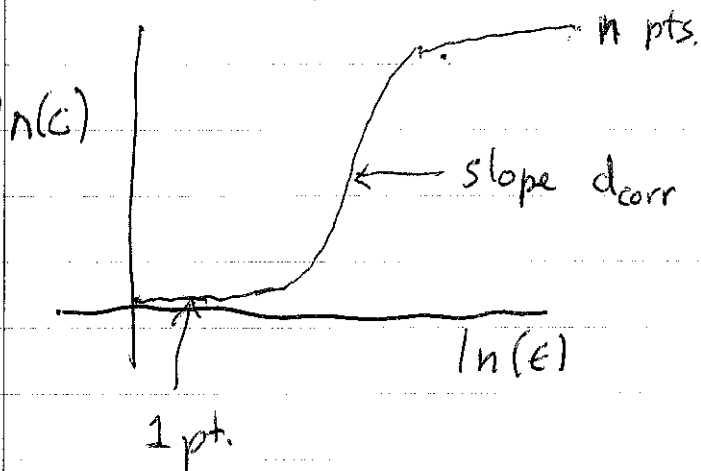


$$u = (x, y, z)$$

Numerically: $\{u_i, i=1, 2, \dots, n\}$ n pts.

consider ball of radius ϵ centered at a pt U , count # of pts. in set inside ball. Expect # of pts. to scale with dim. of attractor, i.e. $N_U(\epsilon) \propto \epsilon^d$, a pointwise estimate of d at U .

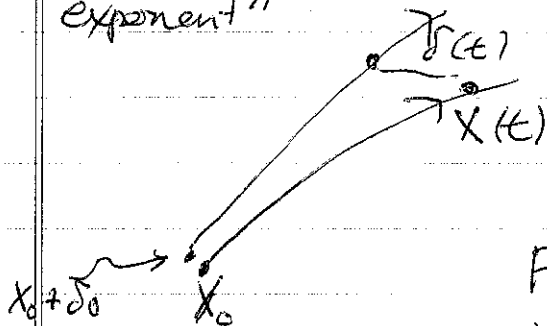
Average over the attractor to obtain "correlation dimension" $C(\epsilon) \propto \epsilon^d$



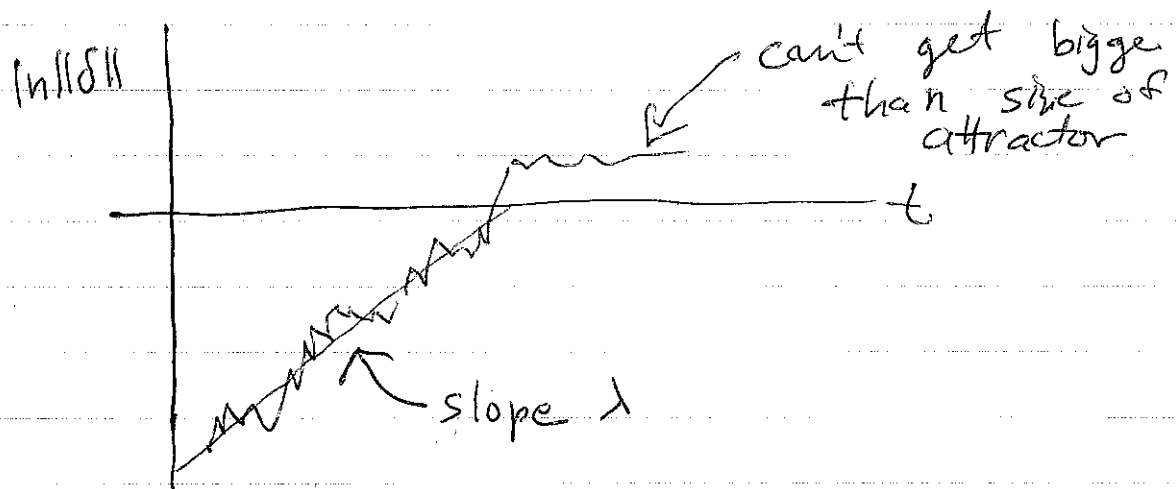
Lorenz with
 $r=28, \sigma=10, b=8/3$

$$d_{\text{corr}} = 2.05 \pm 0.01$$

Hall mark of chaos: "sensitive dependence on initial conditions" — measured by largest "Lyapunov exponent"



Find, initially, $\|\delta(t)\| \sim \|\delta_0\| e^{\lambda t}$
 λ = largest Lyapunov exponent
 $\lambda > 0$ for chaos



Say you know X at time T to 2 decimal places (e.g. $\|\delta(T)\| = 1$) and you want to double to $2T$. How much do you need to improve precision of X_0 ?

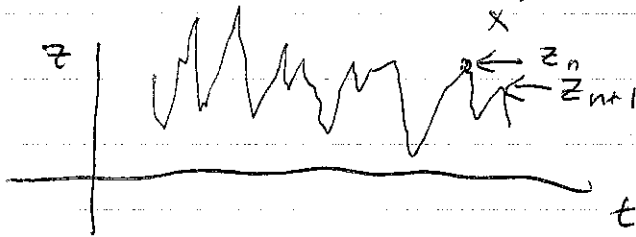
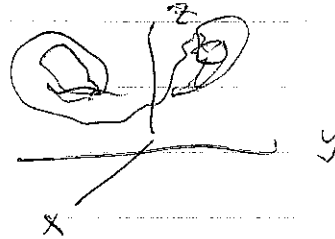
$$\left. \begin{aligned} \|\delta(T)\| = 1 &= e^{\lambda T} \|\delta_0\| \\ \|\delta(2T)\| = 1 &= e^{\lambda 2T} \|\hat{\delta}_0\| \end{aligned} \right\} \begin{aligned} e^{\lambda T} &= 1/\|\delta_0\| \\ e^{2\lambda T} \|\hat{\delta}_0\| &= \frac{\|\hat{\delta}_0\|}{\|\delta_0\|^2} = 1 \end{aligned}$$

$$\|\hat{\delta}_0\| = \|\delta_0\|^2$$

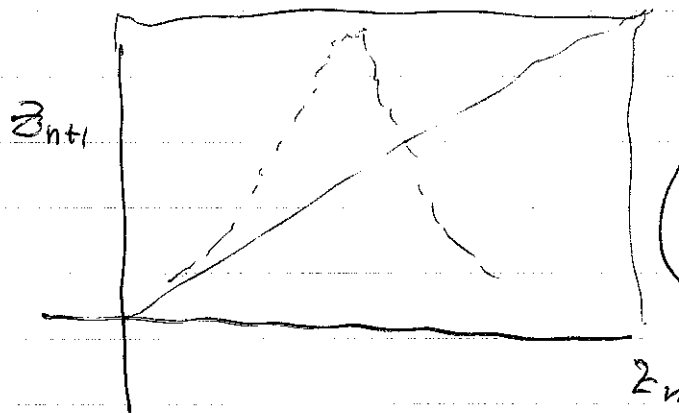
$10^{-10} \rightarrow \sim 10^{-5}$

need to double # of digits.

Lorenz map :



some order
to the chaos



← Lorenz map
(note $f'(z) > 1$
everywhere)

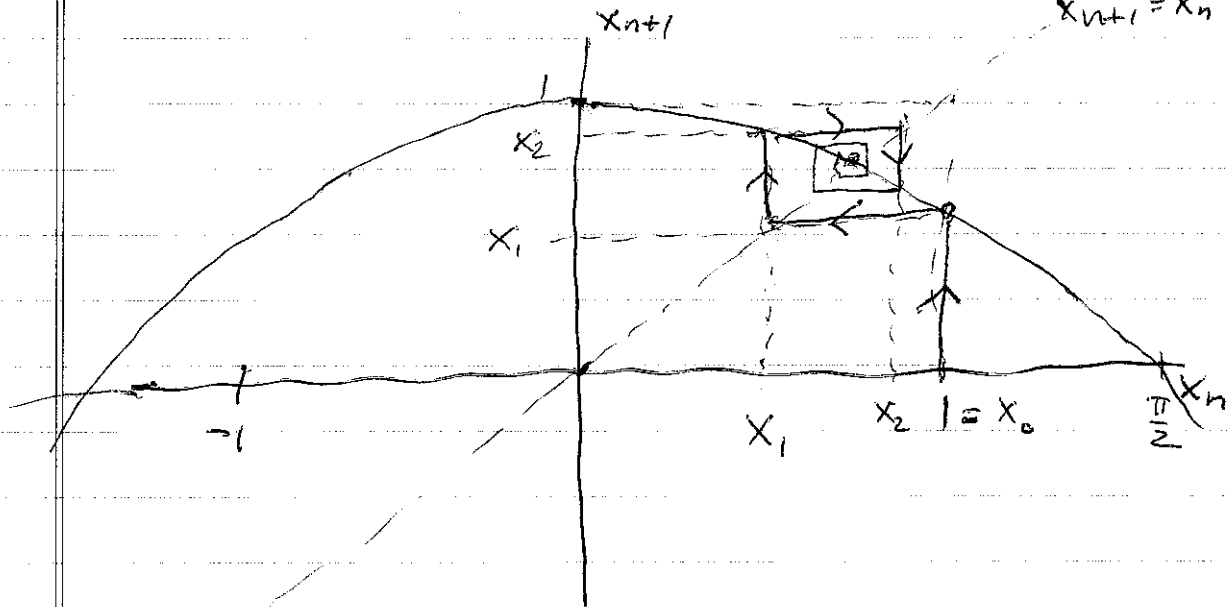
- Lorenz map has some thickness
(not a 1-d curve)

• If there is a ~~limit~~ limit cycle then

Last Topic: 1-d maps & "period-doubling cascade"

$$x_{n+1} = f(x_n)$$

example 1: $x_{n+1} = \cos(x_n)$, $x_n \in [-1, 1]$, $n=1, 2, \dots$
 $x_{n+1} = x_n$



$$x^*: x_{n+1}^* = x_n^*, \quad x^* = \cos(x^*) \\ = 0.739085 \dots$$

Claim: no matter what # you start with in calculator, if you keep pushing cos (in radians), you get to this #. x^* eventually

Linear stability calculation for a fixed pt.
 x^* of a map

$$x^* = f(x^*)$$

fixed-pt eqn.

let $x_n = x^* + \delta_n$

$$x_{n+1} = f(x^* + \delta_n)$$

$$x^* + \delta_{n+1} = \underbrace{f(x^*)}_{x^*} + f'(x^*)\delta_n + \dots$$

$$\delta_{n+1} \approx \underbrace{f'(x^*)}_{\mu} \delta_n \quad \text{is linearized eqn.}$$

$$\delta_0 \rightarrow \mu \delta_0 \rightarrow \mu^2 \delta_0 \rightarrow \mu^3 \delta_0 \rightarrow \dots$$

$$\delta_n = \mu^n \delta_0$$

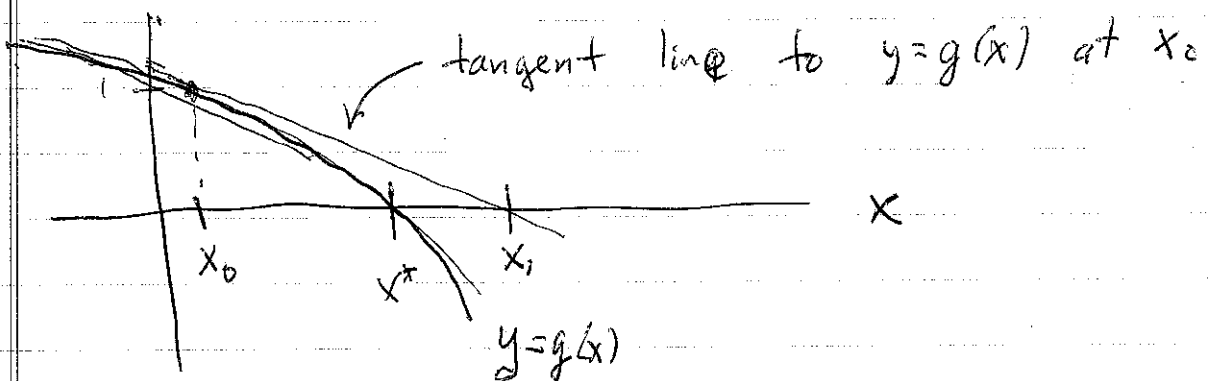
$$\lim_{n \rightarrow \infty} |\delta_n| = 0 \quad \text{if} \quad |\mu| < 1$$

example $x_{n+1} = \cos(x_n) = f(x_n)$

$$f'(x_0^*) = -\sin(x^*) \Rightarrow |\mu| < 1$$

Another example :

Newton's method for finding the zero of a function: Find x^* s.t. $g(x^*) = 0$
 e.g. $(\cos(x) - x = g(x))$



initial guess x_0 , check $g(x_0) \approx 0$?

if not close enough to zero then next guess is x_1

$$g'(x_0) = \frac{g(x_0)}{x_0 - x_1} \Rightarrow (x_0 - x_1) = \frac{g(x_0)}{g'(x_0)}$$

$$x_1 = x_0 - \frac{g(x_0)}{g'(x_0)}, \text{ keep repeating } \dots$$

$$x_{n+1} = x_n - \frac{g(x_n)}{g'(x_n)} = F(x_n)$$

if $g(x) = \cos(x) - x$
 $g'(x) = -\sin(x) - 1$

$$X_{n+1} = X_n + \frac{\cos(x_n) - x_n}{1 + \sin(x_n)} = F(x_n)$$

$n=0$	x_n	$\cos(x_n)$	x_n	$F(x_n)$
0	1	0.540302	1	0.750364
1	←	0.8575..	2	0.739113
2	←	0.6543..	3	0.739085
3	←	0.7935..		
4	←	0.7014..		
5	←	0.764..		

$$x^* = 0.739085, \dots$$

Converges super fast x^* is a super stable fixed pt. of the Newton iteration map.

$$x^* = F(x^*)$$

$$F(x) = x - \frac{g(x)}{g'(x)}$$

$$\mu = F'(x^*) = 1 - \frac{g'(x^*)}{g'(x^*)} + \frac{g(x^*)}{(g'(x^*))^2} g''(x^*)$$

$$= \frac{g(x^*)}{g'(x^*)^2} g''(x^*)$$

but $g(x^*) = 0$
 $\Rightarrow \mu = 0.$